

Tezpur University
Autumn Semester Mid examination-2025.

Course Code: CS519*Full marks=30***Course Title: Computer Networks***Time = 1.5 hours*

(The figures in the right-hand margin indicate full marks for the questions)

****All questions are compulsory**

Q1.	Deduce an expression to find the MAC efficiency of an Ethernet medium. Relate the followings with the MAC efficiency: a. Delay-bandwidth product, b. Propagation and transmission delay.	3+1+1
Q2.	Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.	2
Q3.	Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted over this link to maximally pack them in transit (within the link). What is the minimum number of bits that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.	2
Q4.	What is the required relation between frame time and propagation delay for CSMA-CD? What are the major operations performed under CD?	1+2
Q5.	Which MAC algorithm is the most suitable for satellite communications?	1
Q6.	An IP packet of 6000 bytes in size is transmitted from a HTTP server to the client with no QoS. Client which uses an Ethernet device in layer 2, receives the HTTP data. Show all the L2 and L3 header fields with feasible values for all the packets (unfragmented or fragmented) involved in the communication- a. At the sender side (HTTP Server) b. At the received side (HTTP Client)	2+8
Q7.	How do you identify a unicast MAC address? What do you mean by the notations - a. 10BaseT b. 1000Base-LX	1+2+2
Q8.	How does NAT operate?	2